SOAL 8 (NPM: 91124082)
Solusi dengan Metode Simpleks
Z = 746,15
X1 = 5,38
X2 = 2,31

JAWABAN
Iterasi 1
Subset 1
Maximize Z = 100X1 + 90X2
Subject to
10X1 + 7X2 ≤ 70
5X1 + 10X2 ≤ 50
X2 ≥ 3

Perhitungan subset 1

- **Mencari X1**
  10X1 + 7X2 = 70
  10X1 + 7(3) = 70
  10X1 + 21 = 70
  10X1 = 49
  X1 = 4,9

  5X1 + 10X2 = 50
  5X1 + 10(3) = 50
  5X1 + 30 = 50
  5X1 = 20
  X1 = 4

- **Mencari Z**
  Z = 100X1 + 90X2
  Z = 100(4,9) + 90(3)
  Z = 490 + 270
  Z = 760 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

  Z = 100X1 + 90X2
  Z = 100(4) + 90(3)
  Z = 400 + 270
  Z = 670 (Calon Solusi)

Subset 2
Maximize Z = 100X1 + 90X2
Subject to
10X1 + 7X2 ≤ 70
5X1 + 10X2 ≤ 50
X2 ≤ 2

Perhitungan subset 2

- **Mencari X1**
  10X1 + 7X2 = 70
  10X1 + 7(2) = 70
  10X1 + 14 = 70
  10X1 = 56
  X1 = 5,6

  5X1 + 10X2 = 50
  5X1 + 10(2) = 50
  5X1 + 20 = 50
  5X1 = 30
  X1 = 6

- **Mencari Z**
  Z = 100X1 + 90X2
  Z = 100(5,6) + 90(2)
  Z = 560 + 180
  Z = 740

  Z = 100X1 + 90X2
  Z = 100(6) + 90(2)
  Z = 600 + 180
  Z = 780 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Iterasi 2

Subset 3

Maximize $Z = 100X1 + 90X2$

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X1 \geq 6}$

Perhitungan subset 3

- **Mencari X2**

  $10X1 + 7X2 = 70$

  $10(6) + 7X2 = 70$

  $60 + 7X2 = 70$

  $7X2 = 10$

  $\mathbf{X2 = \frac{10}{7} = 1{,}43}$

  $5X1 + 10X2 = 50$

  $5(6) + 10X2 = 50$

  $30 + 10X2 = 50$

  $10X2 = 20$

  $X2 = 2$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(6) + 90(1{,}43)$

  $Z = 600 + 128{,}7$

  $\mathbf{Z = 728{,}7}$

  $Z = 100X1 + 90X2$

  $Z = 100(6) + 90(2)$

  $Z = 600 + 180$

  $Z = 780$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Subset 4

Maximize $Z = 100X1 + 90X2$

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X1 \leq 5}$

Perhitungan subset 4

- **Mencari X2**

  $10X1 + 7X2 = 70$

  $10(5) + 7X2 = 70$

  $50 + 7X2 = 70$

  $7X2 = 20$

  $X2 = \frac{20}{7} = 2{,}86$

  $5X1 + 10X2 = 50$

  $5(5) + 10X2 = 50$

  $25 + 10X2 = 50$

  $10X2 = 25$

  $\mathbf{X2 = 2{,}5}$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(5) + 90(2{,}86)$

  $Z = 500 + 257{,}4$

  $Z = 757{,}4$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

  $Z = 100X1 + 90X2$

  $Z = 100(5) + 90(2{,}5)$

  $Z = 500 + 225$

  $\mathbf{Z = 725}$

Iterasi 3

Subset 5

Maximize Z = 100X1 + 90X2

Subject to

10X1 + 7X2 ≤ 70

5X1 + 10X2 ≤ 50

X2 ≥ 2

Perhitungan subset 5

- **Mencari X1**

  10X1 + 7X2 = 70

  10X1 + 7(2) = 70

  10X1 + 14 = 70

  10X1 = 56

  X1 = 5,6

  5X1 + 10X2 = 50

  5X1 + 10(2) = 50

  5X1 + 20 = 50

  5X1 = 30

  X1 = 6

- **Mencari Z**

  Z = 100X1 + 90X2

  Z = 100(5,6) + 90(2)

  Z = 560 + 180

  Z = 740

  Z = 100X1 + 90X2

  Z = 100(6) + 90(2)

  Z = 600 + 180

  Z = 780 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Subset 6 (Tidak Digunakan)

Maximize Z = 100X1 + 90X2

Subject to

10X1 + 7X2 ≤ 70

5X1 + 10X2 ≤ 50

X2 ≤ 1

Perhitungan subset 6

- **Mencari X1**

  10X1 + 7X2 = 70

  10X1 + 7(1) = 70

  10X1 + 7 = 70

  10X1 = 63

  X1 = 6,3

  5X1 + 10X2 = 50

  5X1 + 10(1) = 50

  5X1 + 10 = 50

  5X1 = 40

  X1 = 8

- **Mencari Z**

  Z = 100X1 + 90X2

  Z = 100(6,3) + 90(2)

  Z = 630 + 180

  Z = 810 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

  Z = 100X1 + 90X2

  Z = 100(8) + 90(2)

  Z = 800 + 180

  Z = 980 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Iterasi 4

Subset 7

Maximize $Z = 100X1 + 90X2$

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X2 \geq 3}$

Perhitungan subset 7

- **Mencari X1**

  $10X1 + 7X2 = 70$

  $10X1 + 7(3) = 70$

  $10X1 + 21 = 70$

  $10X1 = 49$

  $X1 = 4{,}9$

  $5X1 + 10X2 = 50$

  $5X1 + 10(3) = 50$

  $5X1 + 30 = 50$

  $5X1 = 20$

  $\mathbf{X1 = 4}$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(4{,}9) + 90(3)$

  $Z = 490 + 270$

  $Z = 760$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

  $Z = 100X1 + 90X2$

  $Z = 100(4) + 90(3)$

  $Z = 400 + 270$

  $\mathbf{Z = 670}$ **(Calon Solusi)**

Subset 8

Maximize $Z = 100X1 + 90X2$

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X2 \leq 2}$

Perhitungan subset 8

- **Mencari X1**

  $10X1 + 7X2 = 70$

  $10X1 + 7(2) = 70$

  $10X1 + 14 = 70$

  $10X1 = 56$

  $\mathbf{X1 = 5{,}6}$

  $5X1 + 10X2 = 50$

  $5X1 + 10(2) = 50$

  $5X1 + 20 = 50$

  $5X1 = 30$

  $X1 = 6$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(5{,}6) + 90(2)$

  $Z = 560 + 180$

  $\mathbf{Z = 740}$

  $Z = 100X1 + 90X2$

  $Z = 100(6) + 90(2)$

  $Z = 600 + 180$

  $Z = 780$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Iterasi 5

Subset 9

Maximize Z = 100X1 + 90X2

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X1 \geq 6}$

Perhitungan subset 8

- **Mencari X2**

  $10X1 + 7X2 = 70$

  $10(6) + 7X2 = 70$

  $60 + 7X2 = 70$

  $7X2 = 10$

  $\mathbf{X2 = \frac{10}{7} = 1{,}43}$

  $5X1 + 10X2 = 50$

  $5(6) + 10X2 = 50$

  $30 + 10X2 = 50$

  $10X2 = 20$

  $X2 = 2$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(6) + 90(1{,}43)$

  $Z = 600 + 128{,}7$

  $\mathbf{Z = 728{,}7}$

  $Z = 100X1 + 90X2$

  $Z = 100(6) + 90(2)$

  $Z = 600 + 180$

  $Z = 780$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Subset 10

Maximize Z = 100X1 + 90X2

Subject to

$10X1 + 7X2 \leq 70$

$5X1 + 10X2 \leq 50$

$\mathbf{X1 \leq 5}$

Perhitungan subset 9

- **Mencari X2**

  $10X1 + 7X2 = 70$

  $10(5) + 7X2 = 70$

  $50 + 7X2 = 70$

  $7X2 = 20$

  $X2 = \frac{20}{7} = 2{,}86$

  $5X1 + 10X2 = 50$

  $5(5) + 10X2 = 50$

  $25 + 10X2 = 50$

  $10X2 = 25$

  $\mathbf{X2 = 2{,}5}$

- **Mencari Z**

  $Z = 100X1 + 90X2$

  $Z = 100(5) + 90(2{,}86)$

  $Z = 500 + 257{,}4$

  $Z = 757{,}4$ → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

  $Z = 100X1 + 90X2$

  $Z = 100(5) + 90(2{,}5)$

  $Z = 500 + 225$

  $\mathbf{Z = 725}$